

SALES ANALYTICS DASHBOARD

Internship Task 2 — Project Documentation

Student

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Program

B.E./B.Tech. in Computer Engineering

Internship

Sumerix Global — Data Science Internship

End-to-end sales analysis using Python, PostgreSQL, SQL, Power BI and DAX.

1. Project Overview

This project was completed as part of Sumerix Global Data Science Internship — Task 2. It analyzes transactional sales data and converts it into KPIs, comparisons, trends, profitability measures and a Power BI dashboard.

2. Objectives

- Analyze overall sales, profit, orders and quantity.
- Compare performance across customer segments and regions.
- Analyze category-level sales, profit and profit margin.
- Study yearly, quarterly and monthly sales performance.
- Perform structured analysis using Python, SQL and PostgreSQL.
- Create a professional Power BI dashboard and DAX measures.

3. Dataset Description

The cleaned dataset contains order, customer, product, geographic, financial and time-related information.

Fields: Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Region, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit, Order Year, Order Month, Order Month Name, Order Quarter, Order Day, Order Weekday and Shipping Days.

4. Tools & Technologies

Tool	Purpose
Python	Data analysis
Pandas / NumPy	Data manipulation and numerical analysis
Matplotlib / Seaborn	Visualization
Jupyter Notebook	Exploratory analysis
PostgreSQL	Database storage and SQL analysis
SQL	Business queries and aggregations
Power BI	Dashboard and visualization
DAX	Dynamic measures and calculations

5. Project Workflow

Raw Dataset → Data Preparation → PostgreSQL → SQL Analysis → Python/Jupyter Analysis → Power BI → DAX → Dashboard → Documentation

6. Data Preparation

- Reviewed the dataset structure and required fields.
- Prepared date-related fields for time-based analysis.
- Used Order Year, Order Month, Order Month Name and Order Quarter for trend analysis.
- Calculated Shipping Days for shipping-related analysis.
- Prepared the cleaned dataset for PostgreSQL, Python and Power BI.

7. SQL & PostgreSQL Analysis

PostgreSQL was used to store the cleaned sales data and perform structured SQL analysis.

- KPI calculations: sales, profit, orders, quantity and average order value.
- Sales and profit by segment and region.
- Category sales, profit and profit margin.
- Yearly sales and year-over-year growth.
- Quarterly sales, profit and profit margin.
- Customer and product analysis.
- Monthly and trend-based analysis.

8. KPI Summary

KPI	Value
Total Sales	₹12.51M
Total Profit	₹2.24M
Total Orders	5,000
Total Quantity	27,507
Average Order Value	≈ ₹2,501.36

9. Segment Analysis

Segment	Sales	Profit
Corporate	₹4,244,266.85	₹732,570.05
Home Office	₹4,136,313.47	₹775,561.84
Consumer	₹4,126,235.19	₹731,579.24

10. Region Analysis

Region	Sales	Profit
South	₹3,247,440.79	₹574,301.38
West	₹3,175,737.09	₹569,469.81
Central	₹3,084,820.38	₹558,202.61
East	₹2,998,817.25	₹537,737.33

11. Category Profit Margin

Category	Sales	Profit	Margin
Furniture	₹4,169,553.39	₹753,823.28	18.079%
Office Supplies	₹4,184,882.48	₹754,427.39	18.027%
Technology	₹4,152,379.64	₹731,460.46	17.615%

12. Yearly Sales Analysis

Year	Sales	Growth
2023	₹4,600,381.57	N/A
2024	₹6,171,980.63	34.162%
2025	₹1,734,453.31	-71.898%

13. Quarterly Analysis

Quarter	Sales	Profit	Margin
Q1	₹3,004,572.67	₹524,372.55	17.452%
Q2	₹3,185,401.16	₹574,273.43	18.028%
Q3	₹3,157,588.84	₹560,533.18	17.752%
Q4	₹3,159,252.84	₹580,531.97	18.376%

14. Power BI Dashboard

The dashboard presents KPI cards and visual analysis including sales by category and region, monthly sales trend, profit by category and region, customer-segment analysis, quarterly sales and profit, top products by profit and top customers by sales.

- Total Sales
- Total Profit
- Total Orders
- Total Quantity
- Average Order Value
- Sales by Category
- Sales by Region
- Monthly Sales Trend
- Profit by Category
- Sales by Customer Segment
- Profit by Region
- Quarterly Sales & Profit
- Top 10 Products by Profit
- Top 10 Customers by Sales

15. DAX Measures

DAX was used in Power BI to create dynamic measures and analytical calculations.

Total Sales

Total Sales = SUM('public sales'[sales])

Total Profit

Total Profit = SUM('public sales'[profit])

Total Orders

Total Orders = DISTINCTCOUNT('public sales'[order_id])

Total Quantity

Total Quantity = SUM('public sales'[quantity])

Average Order Value

Average Order Value = DIVIDE([Total Sales], [Total Orders])

Profit Margin

Profit Margin = DIVIDE([Total Profit], [Total Sales]) * 100

Year-over-Year Growth

YoY Growth =
VAR CurrentYear = [Total Sales]
VAR PreviousYear = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('public sales'[order_date]))
RETURN DIVIDE(CurrentYear - PreviousYear, PreviousYear) * 100

16. Visualizations

The notebook visualizations were extracted into a separate PDF named Sales_Analytics_Visualizations.pdf for storage under the visualizations/ folder.

17. Project Structure

```
sales-bi-dashboard/  
├─ datasets/  
├─ database/  
├─ notebooks/  
├─ scripts/  
├─ dashboard/  
├─ reports/  
├─ visualizations/  
├─ documentation/  
├─ requirements_task2.txt  
├─ README.md  
└─ .gitignore
```

18. Deliverables

- Cleaned dataset
- PostgreSQL database / related files
- Jupyter Notebook
- SQL analysis script

- Power BI dashboard (.pbix)
- Sales Analytics Visualizations PDF
- Task 2 project report
- Project documentation PDF
- README.md
- requirements_task2.txt
- .gitignore

19. Conclusion

The project demonstrates an end-to-end sales analytics workflow from data preparation and database analysis to visualization and dashboard development. Python, SQL, PostgreSQL, Power BI and DAX were combined to produce structured KPIs, trend analysis, profitability analysis and business-oriented visualizations.

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